

संभव BATCH - CRASH COURSE

Hard Level: Interior of Earth & Distribution of Oceans and Continents

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SSC CGL Tier-II Pattern | Multi-statement & Analytical MCQs | No Answer Key

Q1. Consider the following statements regarding the behavior of seismic waves in the Earth's interior:

- I. P-waves travel faster through solid materials than through liquids.
- II. S-waves are transverse waves that create a shadow zone between 105° and 145° from the epicenter.
- III. The emergence of P-waves in the shadow zone beyond 145° proves the existence of a solid inner core.

Which of the statements given above is/are NOT correct?

- (A) I and II only
- (B) II only
- (C) II and III only
- (D) III only

SSC CGL Tier-II 2022 Pattern

Q2. Assertion (A):

The Earth's magnetic field is generated by the geodynamo effect in the solid inner core.

Reason (R): The inner core is composed primarily of an iron-nickel alloy under extreme pressure, preventing it from melting despite high temperatures.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

SSC CPO 2021 Pattern

Q3. Match List-I (Seismic Discontinuities) with List-II (Earth's Layers):

List-I	List-II
A. Conrad Discontinuity	1. Between upper and lower mantle
B. Repetti Discontinuity	2. Between crust and upper mantle
C. Mohorovicic Discontinuity	3. Between outer and inner core
D. Lehmann Discontinuity	4. Between upper and lower crust

Codes:

- (A) A-4, B-1, C-2, D-3
(B) A-2, B-3, C-4, D-1
(C) A-4, B-3, C-2, D-1
(D) A-2, B-1, C-4, D-3

SSC Selection Post Phase XI

Q4. Statement I:

The oceanic crust is significantly younger than the continental crust.

Statement II: The continuous process of subduction at convergent boundaries destroys older oceanic crust, while new crust is formed at mid-oceanic ridges.

Which of the following is correct?

- (A) Statement I is correct, Statement II is incorrect.
(B) Statement I is incorrect, Statement II is correct.
(C) Both statements are correct.
(D) Both statements are incorrect.

SSC CGL 2023

Q5. Regarding the internal structure of the Earth, which of the following statements is/are correct?

1. The lithosphere has a thickness ranging from 10 to 200 km.
2. The asthenosphere is entirely in a liquid state, allowing tectonic plates to float freely.
3. The density of the Earth's core is approximately 13 g/cm^3 .

Select the correct answer:

- (A) 1 and 2 only
(B) 1 and 3 only
(C) 2 and 3 only
(D) 1, 2 and 3

SSC CGL Tier-II 2021

Q6. Assertion (A):

Arthur Holmes proposed the Convection Current Theory in the 1930s to explain the mechanism behind continental drift.

Reason (R): He argued that radioactive elements in the mantle generate thermal differences, creating convection currents that drive the lithospheric plates.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

SSC CHSL 2022 Pattern

Q7. Which of the following is NOT an argument originally presented by Alfred Wegener in support of the Continental Drift Theory?

- (A) Jig-Saw-Fit of the shorelines of South America and Africa.
- (B) Paleomagnetic striping on the ocean floor proving magnetic reversals.
- (C) Occurrence of rich placer deposits of gold on the Ghana coast without a local source rock.
- (D) Distribution of Glossopteris flora across India, South America, and Antarctica.

SSC MTS 2021

Q8. Match List-I (Minor Tectonic Plates) with List-II (Geographical Location):

List-I	List-II
A. Scotia Plate	1. Between Philippine and Indian plates
B. Cocos Plate	2. Between South America and Antarctica
C. Caroline Plate	3. Between Central America and Pacific plate
D. Nazca Plate	4. Between South America and Pacific plate

Codes:

- (A) A-2, B-3, C-1, D-4
- (B) A-4, B-1, C-2, D-3
- (C) A-2, B-1, C-3, D-4
- (D) A-4, B-3, C-1, D-2

SSC CPO 2022

Q9. Consider the following statements about Paleomagnetism:

- I. Rocks forming at mid-oceanic ridges record the Earth's magnetic polarity at the time of their cooling.
- II. Magnetic stripes on the ocean floor are asymmetrical on either side of the mid-oceanic ridges.
- III. It provided the ultimate definitive evidence for Harry Hess's Sea Floor Spreading theory.

Which of the statements given above is/are NOT correct?

- (A) I and II only
- (B) II only
- (C) I and III only
- (D) III only

SSC CGL Tier-II 2023

Q10. Assertion (A):

Deep-focus earthquakes are exclusively associated with divergent plate boundaries.

Reason (R): At divergent boundaries, the sinking of a cold, dense lithospheric slab deep into the asthenosphere generates intense seismic activity.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) Both A and R are false.

SSC Selection Post Phase X

Q11. Which of the following statements regarding the Earth's mantle is INCORRECT?

- (A) It extends from Mohorovicic discontinuity to a depth of 2,900 km.
- (B) The upper portion of the mantle is called the asthenosphere.
- (C) It constitutes approximately 83% of the Earth's volume.
- (D) The lower mantle exists in a highly viscous liquid state due to extreme temperatures.

SSC CGL 2021 Pattern

Q12. The concept of 'Polar Wandering' refers to:

- (A) The physical movement of the geographic poles over geological time.
- (B) The apparent movement of the magnetic poles as deduced from paleomagnetic data, confirming continental drift.
- (C) The reversal of the Earth's magnetic field every few million years.
- (D) The shifting of the Earth's axis of rotation due to Milankovitch cycles.

SSC CHSL 2021

Q13. Match List-I (Evidence of Continental Drift) with List-II (Specific detail):

List-I	List-II
A. Tillite	1. Gold deposits lacking source rock
B. Placer Deposits	2. Mesozoic reptile found in SA and Africa
C. Mesosaurus	3. Sedimentary rock formed out of glacial deposits
D. Glossopteris	4. Fossil fern found across Gondwanaland

Codes:

(A) A-3, B-1, C-2, D-4

(B) A-1, B-3, C-4, D-2

(C) A-3, B-2, C-1, D-4

(D) A-2, B-1, C-4, D-3

SSC CPO 2020

Q14. Statement I:

In an earthquake, P-waves cause particles to vibrate perpendicular to the direction of wave propagation.

Statement II: S-waves cause particles to vibrate parallel to the direction of wave propagation, creating crests and troughs.

Which of the following is correct?

(A) Statement I is correct, Statement II is incorrect.

(B) Statement I is incorrect, Statement II is correct.

(C) Both statements are correct.

(D) Both statements are incorrect.

SSC CGL Tier-II 2020

Q15. Consider the following geological features:

1. Deep ocean trenches
2. Volcanic island arcs
3. Mid-oceanic ridges
4. Fold mountains

Which of the features given above are typically formed at a convergent plate boundary?

(A) 1, 2 and 3 only

(B) 1, 2 and 4 only

(C) 2, 3 and 4 only

(D) 1, 3 and 4 only

SSC MTS 2022

Q16. Assertion (A):

The Himalayas continue to increase in height today.

Reason (R): The Eurasian plate is actively subducting beneath the Indo-Australian plate, creating massive volcanic uplifts in the Tibetan plateau.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

SSC CGL 2022

Q17. Which of the following forces did Alfred Wegener propose as the driving mechanism for continental drift, which was later rejected by modern geologists as insufficient?

- (A) Mantle convection currents
- (B) Ridge push and slab pull
- (C) Pole-fleeing force and tidal force
- (D) Isostatic adjustment

SSC CHSL 2020

Q18. Consider the following statements regarding the Earth's core:

- I. The outer core behaves like a liquid, which is essential for the generation of the Earth's magnetic field.
- II. The Gutenberg discontinuity separates the inner core from the outer core.
- III. The density of the core ranges between 5 g/cm^3 to 13 g/cm^3 .

Which of the statements given above is/are NOT correct?

- (A) I and II only
- (B) II only
- (C) II and III only
- (D) I and III only

SSC CPO 2023

Q19. Match List-I (Plate Boundaries) with List-II (Associated Features):

List-I	List-II
A. Divergent Boundary	1. San Andreas Fault
B. Convergent Boundary	2. Mid-Atlantic Ridge
C. Transform Boundary	3. Mariana Trench

Codes:

- (A) A-2, B-3, C-1
- (B) A-3, B-2, C-1
- (C) A-2, B-1, C-3
- (D) A-1, B-3, C-2

SSC CGL Tier-II 2021

Q20. Assertion (A):

The age of the rocks on the ocean floor increases as one moves away from the crest of the mid-oceanic ridges.

Reason (R): New oceanic crust is continuously formed at the ridges due to the upwelling of magma and is pushed laterally outwards.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

SSC Selection Post Phase XI

Q21. Statement I:

The Ring of Fire in the Pacific Ocean is a direct result of divergent plate boundaries.

Statement II: It contains over 75% of the world's active and dormant volcanoes.

Which of the following is correct?

- (A) Statement I is correct, Statement II is incorrect.
- (B) Statement I is incorrect, Statement II is correct.
- (C) Both statements are correct.
- (D) Both statements are incorrect.

SSC CGL 2021

Q22. Which of the following is/are considered direct sources of information about the interior of the earth?

1. Mining and deep ocean drilling projects
2. Volcanic eruptions
3. Seismic waves
4. Magnetic surveys

Select the correct answer using the codes below:

- (A) 1 and 2 only
- (B) 3 and 4 only
- (C) 1, 2 and 3 only
- (D) 1, 2, 3 and 4

SSC CHSL 2022 Pattern

Q23. In the context of plate tectonics, the term "triple junction" refers to:

- (A) A point where a continental plate, an oceanic plate, and the asthenosphere meet.
- (B) The point where the boundaries of three tectonic plates meet.
- (C) The interaction zone of P-waves, S-waves, and L-waves.
- (D) The geological boundary separating the crust, mantle, and core.

SSC CPO 2021

Q24. Consider the following statements about the distribution of earthquakes:

- I. The Alpine-Himalayan belt accounts for about 15% of all earthquakes worldwide.
- II. Earthquakes at mid-oceanic ridges are generally of deep focus.
- III. The Circum-Pacific belt is associated with the subduction of oceanic lithosphere.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

SSC CGL Tier-II 2022

Q25. Match List-I (Tectonic Plate) with List-II (Type of Plate):

List-I	List-II
A. Pacific Plate	1. Mostly Continental
B. Eurasian Plate	2. Mostly Oceanic
C. Indo-Australian Plate	3. Minor Plate
D. Scotia Plate	4. Continental and Oceanic

Codes:

(A) A-2, B-1, C-4, D-3

(B) A-2, B-4, C-1, D-3

(C) A-4, B-1, C-2, D-3

(D) A-2, B-1, C-3, D-4

SSC MTS 2021

Q26. Assertion (A):

Rayleigh waves and Love waves travel only along the surface of the Earth.

Reason (R): They are generated by the interaction of body waves with the surface rocks and lose energy rapidly with depth.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

SSC Selection Post 2022

Q27. According to Harry Hess's concept of sea-floor spreading, what happens to the older oceanic crust?

(A) It gets folded into high mountain ranges.

(B) It gets subducted into the mantle at oceanic trenches and melts.

(C) It turns into continental crust over billions of years.

(D) It dissolves directly into the ocean water.

SSC CGL 2023

Q28. Which of the following statements about 'Gondwanaland' is NOT correct?

(A) It was the southern part of the supercontinent Pangaea.

(B) It included present-day South America, Africa, India, Antarctica, and Australia.

(C) It broke apart due to the collision with the Eurasian plate.

(D) The Tethys Sea separated it from Laurasia.

SSC CPO 2022

Q29. Statement I:

The Earth's crust is thickest beneath major mountain ranges like the Himalayas.

Statement II: The principle of isostasy requires that higher topographic elevations have deeper crustal 'roots' extending into the mantle.

Which of the following is correct?

- (A) Statement I is correct, Statement II is incorrect.
- (B) Statement I is incorrect, Statement II is correct.
- (C) Both statements are correct.
- (D) Both statements are incorrect.

SSC CHSL 2021

Q30. Match List-I (Earth Layer) with List-II (Dominant Composition):

List-I	List-II
A. Continental Crust	1. Iron and Nickel
B. Oceanic Crust	2. Silicon and Magnesium
C. Core	3. Iron and Silicates
D. Mantle	4. Silicon and Aluminium

Codes:

- (A) A-4, B-2, C-1, D-3
- (B) A-2, B-4, C-1, D-3
- (C) A-4, B-2, C-3, D-1
- (D) A-1, B-2, C-4, D-3

SSC CGL Tier-II 2020

Q31. Consider the following regarding the S-wave shadow zone:

1. It is smaller than the P-wave shadow zone.
2. It exists because S-waves cannot travel through the liquid outer core.
3. It extends entirely from 105° on one side of the epicenter to 105° on the other side.

Which of the statements given above are correct?

- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 and 3 only
- (D) 1, 2 and 3

SSC CGL 2022 Pattern

Q32. Assertion (A):

The rate of increase in temperature with respect to depth (temperature gradient) remains constant from the crust to the core.

Reason (R): The radioactive elements responsible for generating heat are uniformly distributed throughout the Earth's interior.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) Both A and R are false.

SSC Selection Post Phase X

Q33. Which of the following correctly describes a 'Constructive Plate Margin'?

- (A) Plates collide, leading to the formation of fold mountains.
- (B) Plates move away from each other, and new magma wells up to form crust.
- (C) Plates slide past each other without creating or destroying crust.
- (D) One plate is subducted beneath another, creating a trench.

SSC MTS 2022

Q34. The deepest drill ever made by scientists to study the Earth's interior is located in:

- (A) The Mariana Trench
- (B) The Kola Peninsula in Russia
- (C) The San Andreas Fault
- (D) The Mid-Atlantic Ridge

SSC CHSL 2020

Q35. Statement I:

Pangaea was completely assembled by the early Permian period and began to break up around 200 million years ago.

Statement II: The initial break-up of Pangaea formed the Pacific Ocean.

Which of the following is correct?

- (A) Statement I is correct, Statement II is incorrect.
- (B) Statement I is incorrect, Statement II is correct.
- (C) Both statements are correct.
- (D) Both statements are incorrect.

SSC CPO 2021

Q36. Match List-I (Phenomenon) with List-II (Cause):

List-I	List-II
A. Placer deposits in Ghana	1. Found in Antarctica
B. Glacial Tillite in India	2. Base rocks located in Brazil
C. Coal seams	3. Once located near South Pole

Codes:

- (A) A-2, B-3, C-1
- (B) A-3, B-2, C-1
- (C) A-1, B-2, C-3
- (D) A-2, B-1, C-3

SSC CGL Tier-II 2023

Q37. Which of the following is NOT an attribute of the Continental Crust compared to the Oceanic Crust?

- (A) It has a lower average density.
- (B) It is composed primarily of granitic rocks.
- (C) It is older in geological age on average.
- (D) It is much thinner than the oceanic crust.

SSC CGL 2021

Q38. Assertion (A):

Transform faults rarely produce volcanic activity.

Reason (R): At transform boundaries, lithospheric plates slide past one another horizontally without upwelling of magma or subduction.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

SSC CHSL 2022

Q39. Consider the following features:

- I. V-shaped valleys
- II. Deep ocean trenches
- III. Mid-oceanic ridges

Which of the above are exclusively associated with divergent plate boundaries?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

SSC CPO 2023

Q40. What is the approximate percentage of the Earth's total volume that the core accounts for?

- (A) 1%
- (B) 16%
- (C) 83%
- (D) 33%

SSC MTS 2021

Q41. Statement I:

The epicenter is the point inside the Earth where an earthquake rupture originates.

Statement II: The hypocenter is the point directly above it on the surface of the Earth.

Which of the following is correct?

- (A) Statement I is correct, Statement II is incorrect.
- (B) Statement I is incorrect, Statement II is correct.
- (C) Both statements are correct.
- (D) Both statements are incorrect.

SSC CGL Tier-II 2022

Q42. In the context of Paleomagnetism, the term 'Curie Point' refers to:

- (A) The maximum depth to which drilling is possible.
- (B) The temperature above which magnetic materials lose their permanent magnetic properties.
- (C) The exact location where a compass needle points straight down.
- (D) The point of collision between two continental plates.

SSC Selection Post Phase XI

Q43. Match List-I (Mantle Layer) with List-II (State/Characteristic):

List-I	List-II
A. Uppermost Mantle	1. Rigid, part of Lithosphere
B. Asthenosphere	2. Plastic, highly viscous
C. Mesosphere (Lower Mantle)	3. Solid due to immense pressure

Codes:

- (A) A-1, B-2, C-3
- (B) A-2, B-1, C-3
- (C) A-3, B-2, C-1
- (D) A-1, B-3, C-2

SSC CGL 2020 Pattern

Q44. Which of the following theories finally proved that the ocean floor was not an ancient, static surface, but a dynamic and constantly renewing structure?

- (A) Continental Drift Theory
- (B) Convection Current Theory
- (C) Sea-Floor Spreading Theory
- (D) Isostasy

SSC CHSL 2021

Q45. Assertion (A):

The Himalayas do not possess active volcanoes despite being at a plate boundary.

Reason (R): The collision is between two continental plates, and there is no subduction of water-rich oceanic crust to generate magma.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

SSC CPO 2020

Q46. The boundary separating the crust from the mantle was discovered by analyzing:

- (A) Magnetic field reversals
- (B) The sudden increase in the velocity of seismic waves
- (C) Volcanic lava composition
- (D) Meteorite compositions

SSC CGL Tier-I 2023

Q47. Consider the following regarding the Juan de Fuca Plate:

1. It is a minor tectonic plate.
2. It is subducting beneath the northern part of the North American Plate.
3. This subduction is responsible for the Cascade volcanic arc.

Which of the statements given above are correct?

- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 and 3 only
- (D) 1, 2 and 3

SSC MTS 2022

Q48. Statement I:

Isoclines are lines joining places of equal seismic intensity.

Statement II: Isoseismal lines join places that experience earthquake tremors at the exact same time.

Which of the following is correct?

- (A) Statement I is correct, Statement II is incorrect.
- (B) Statement I is incorrect, Statement II is correct.
- (C) Both statements are correct.
- (D) Both statements are incorrect.

SSC CGL Tier-II 2021

Q49. In Wegener's Continental Drift Theory, the term 'Panthalassa' specifically referred to:

- (A) The northern landmass
- (B) The southern landmass
- (C) The universal ocean
- (D) The primeval sea separating Laurasia and Gondwana

SSC CHSL 2019

Q50. Match the Ocean Trenches with their respective Oceans:

Trench	Ocean
A. Mariana Trench	1. Atlantic Ocean
B. Puerto Rico Trench	2. Indian Ocean
C. Sunda Trench	3. Pacific Ocean

Codes:

- (A) A-3, B-1, C-2
(B) A-2, B-3, C-1
(C) A-3, B-2, C-1
(D) A-1, B-2, C-3

SSC CGL 2022